COMPUTER SYSTEMS AND PLATFORM TECHNOLOGIES

IT0047

FEU ALABANG FEU DILIMAN FEU TECH

Technology Driven by Innovation

MODULE 6.1

Introduction to TASM Progamming

At the end of the lesson, the student should be able to :

a. To be able to learn the basics of turbo assembler.

b. To properly assemble and link a program using TASM and TLINK.

c. To be able to create an assembly language program using turbo assembler with SK as editor.

d. Demonstrate and apply the use of the Service Function Call 02H of INT 21H in the assembly language.

- Execute of an assembly language program using service function call 02H, INT 21

Introduction to TASM Progamming

Text Editor

A text editor is a program that allows you to enter and prepare your program from the ordinary keyboard into a computer readable form. It also allows you to save this file into the disk for later use. The assembler and loader program require the inputs to be saved files from the disk. A text editor can be any popular work processor or edit program that can be produce pure ***ASCII*** text. As a standard convention, program written in assembly language are usually given the filename with an extension of ***.ASM***. This is also the default filename extension that ***MASM*** searches.

Introduction to TASM Progamming

Assembler

A program written in assembly language is translated to machine code by an assembler. Assembler and the corresponding assembly language mnemonics are generally limited to use with one particular microprocessor, which limits their portability, or use on other machines. Today's assembler do much more than translate assembly language mnemonics into binary code.

Introduction to TASM Progamming

Loader

Before the microprocessor can execute any machine instruction, it must first be loaded into memory accessible to it. The loader is the program that actually takes the machine instructions (object code) and places it in memory at the specified starting address for execution. Loaders range from the very simple to the very complex.

Introduction to TASM Progamming

Directives – indicates how an operand or section of a program is to be processed by the assembler

- model tiny – uses 64 KB both for the source code and the data code
- model small – uses 64KB each for the source code and data code
- code – segment that holds the program
- org 100h – origin, indicates the start of a program
- start: / end start – indicates the beginning and end of a program or a section of a program

Introduction to TASM Progamming

Display single character

Interrupt 21h, service number 2

dl – register to hold the character to output

ah=2 – service number to output a single character stored in register dh

ex.

```
mov ah,2
mov dl,'A'
int 21h
```

Introduction to TASM Progamming

Terminate the program properly

Interrupt 20h

or

Interrupt 21h, service number 4ch

ah = 4ch – service number to terminate the program properly stored in register ah

ex.

int 20h / mov ah, 4ch

int 21h

MODULE 6.2

Character Input Program

At the end of the lesson, the student should be able to :

- **To be able to compare the three character input services of INT 21h.**
- **The students should be able to create character input and password protected program.**

String Manipulation Tasm Program

Strings can't usually fit in a register, string are then placed in the memory and then pass the address of the string in memory of two of the registers, the segment address in DS and offset address in DX.

The string output sends a string of characters to the standard output.

On entry: AH = 09h.

DS = segment address of the first character of the string
DX = offset address of the first character of the string

Service 9 display string of character starting with the first character (address in DS: DX) output, but not including, the character "$" (ASCII24H)

String Manipulation Tasm Program

- **Int 21h, service 9**
- display string
- ah = 9
- [ds:dx] = address of the string
- ds = address or location of the string
- dx = register to hold the string
    -

String Manipulation Tasm Program

how to declare a string

Syntax :

var_name db “string$”

db = define byte

$ = string terminator

MODULE 6.3

String Manipulation Tasm Program

At the end of the lesson, the student should be able to :

a. To learn how to print strings using service 9 of INT 21h

b. To learn how to process string in assembly language level

c. To apply loop and call & return instruction.

d. To create a program that provides a simple screen output by using the string output service.

Character Input Tasm Program

INT 21 Function 01H: Keyboard Input with Echo. This operation accepts a character from the keyboard buffer or, if none is present, waits for keyboard entry.

INT 21 Function 07H: Direct Keyboard Input without Echo. This peration works like function 01H, except that the entered character does not echo on the screen and the operation does not respond to a Ctrl-Break request. It could be used to key in a password that is to be visible.

Character Input Tasm Program

INT 21 Function 08H: Keyboard Input without Echo. This operation works like function 01H, except that the entered character does not echo on the screen.

Introduction to TASM Progamming

Clear screen

Interrupt 10h, service number 3

ax = 03 – service number to clear the screen stored in register ax

ex.

```
mov ax,03
int 10h
```

Introduction to TASM Progamming

Carriage return

dl=13 – service number for carriage return stored in register dl

ex.

```
mov dl,13
int 21h
```

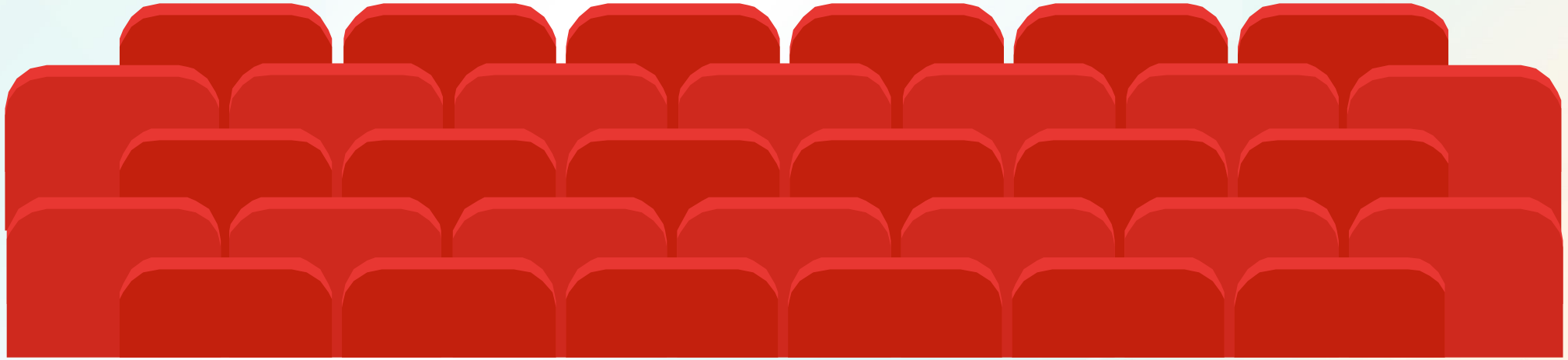
New Line

dl =10 – service number for new line stored in register dl

ex.

```
mov dl,10
int 21h
```

Q&A SESSION

ASK ANY QUESTION RELATED TO OUR TOPIC FOR TODAY.

THANK YOU